CASE STUDY · E-COMMERCE · GROWTH & PRICING ANALYTICS

Two Levers, One Growth Story at OJ Commerce

Customer segmentation that lifted conversion and AOV — paired with a dynamic pricing engine that won the Amazon Buy Box on top-selling SKUs.

+15%
CONVERSION RATE LIFT

+5–7%
AVERAGE ORDER VALUE

+15%
BUY BOX OCCUPANCY

Thousands
OF SKUs MANAGED

Arun · Analytics Lead · OJ Commerce

Multi-channel retail where margin is won by inches, daily.

THE COMPANY

OJ Commerce is a multi-brand e-commerce retailer selling across Amazon, Walmart, and direct-to-consumer sites. Thousands of SKUs, three sales channels, and a constantly-shifting competitive pricing landscape.

MY ROLE

Led analytics and data science for marketplace growth and pricing — partnering with Marketing, Merchandising, and Operations to turn transaction data into commercial decisions.

Amazon

Highest-volume channel. Buy Box occupancy directly drives sales — losing it costs revenue immediately.

Walmart

Growing channel with similar dynamics — pricing-sensitive, marketplace-mediated, competitor-rich.

Direct-to-consumer

Owned-channel where customer relationships and tailored offers compound — different optimization problem.

Two complementary levers — owned customer demand and marketplace supply.

Both projects targeted commercial outcomes, but operated on different sides of the funnel — and reinforced each other.

PROJECT 01

Customer Segmentation

DTC channel · k-means on transactions

WHAT IT DID

- Clustered customers using RFM, product affinity, margin, geography
- Partnered with Marketing & Merchandising on tailored offers
- Lifted conversion ~15%, AOV ~5–7%

PROJECT 02

Buy Box Optimization

Amazon channel · dynamic pricing + inventory

WHAT IT DID

- Built price-elasticity + competitor-monitoring + safety-stock model
- Dynamic pricing engine with inventory gating rules
- Lifted Buy Box occupancy ~15% on top-selling SKUs

Two years of transactions, k-means clustering, tailored offers.

THE PROBLEM

Marketing was running broad campaigns to a customer base with hugely different behaviors — first-time buyers, repeat purchasers, high-margin enthusiasts, deal-seekers. One offer, one message, one playbook didn't fit any of them well.

THE PROPOSAL

Cluster customers in Python on two years of transactions across thousands of SKUs. Use the segments to drive targeted offers, product recommendations, and email cadence.

FEATURE FAMILIES (PER CUSTOMER)

RFM behavior

Recency, frequency, monetary value — the core engagement signal.

Product affinity

Which categories they buy from, repeat-purchase patterns.

Margin

Average product margin — distinguishes premium from deal-seekers.

Geography

Region — proxy for shipping cost, seasonality, channel preference.

PRINCIPLE · *Build features that map to commercial decisions Marketing can act on — not just clusters that look pretty.*

k-means with disciplined feature prep — and a k chosen for usability.

MODELING STEPS

1. **Feature engineering & scaling**

   Built per-customer feature matrix from transaction logs. Log-transformed skewed features (monetary, frequency), then standardized.

2. **Choosing k**

   Elbow + silhouette to narrow the range, then chose a k that produced segments Marketing could act on — not just statistically optimal.

3. **Profiling & naming**

   Profiled clusters by their distinguishing features and gave each a name Marketing could remember and use in campaign briefs.

EXAMPLE SEGMENTS

Illustrative — names disguised, characteristics reflect actual cluster patterns.

Premium loyalists	*High freq + high margin*
Deal seekers	*High freq + low margin*
Lapsing buyers	*Recently active, dropping off*
One-time bargain	*Single low-margin purchase*

KEY CHOICE · *Picked k by what Marketing could actually run — not by silhouette score alone. A 12-cluster optimum nobody could action would have been useless.*

Segments became campaigns. Campaigns moved revenue.

HOW SEGMENTS DROVE ACTION

Tailored promotional offers

Premium loyalists got early access and bundles. Deal-seekers got higher-discount thresholds. One-time bargains got winback offers.

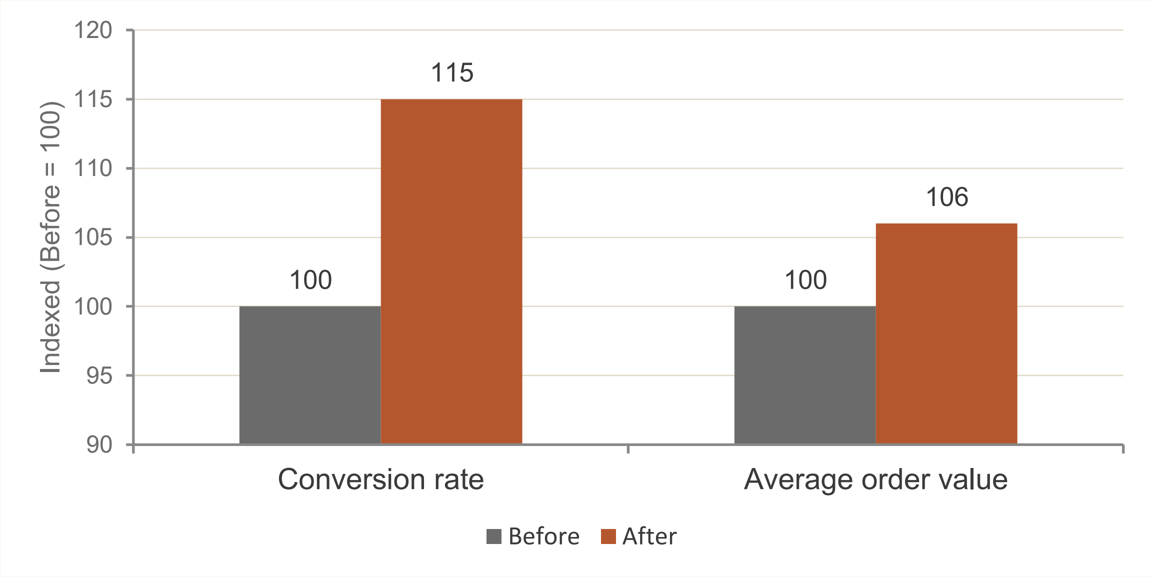
Product recommendations

Recommendation feeds tuned per segment — affinity-driven for repeats, exploration-driven for newer buyers.

Email cadence

Lapsing buyers got reactivation sequences. Loyalists got reduced cadence to avoid fatigue. Each segment, its own pace.

IMPACT — TARGETED vs. PRIOR BROAD CAMPAIGNS



Illustrative — reflects directional gains for confidentiality.

PROJECT 01 IMPACT

+15% conversion rate · +5–7% average order value across DTC campaigns post-segmentation rollout.

When Amazon shoppers click "Add to Cart," 80%+ get the Buy Box winner.

WHAT'S THE BUY BOX?

When a product has multiple sellers on Amazon, only one wins the default "Add to Cart" position. The Buy Box winner gets the overwhelming majority of sales. Lose it, and your offer sits in a list most shoppers never click into.

WHY IT'S COMPLEX

Amazon's algorithm weighs price, fulfillment, inventory level, seller metrics, and more — and competitor prices change constantly. Manual repricing can't keep up at SKU-scale.

THREE FORCES THE MODEL HAD TO BALANCE

Price competitiveness

Lower price helps Buy Box odds — but margin compression hurts the bottom line.

Inventory protection

Selling out of fast-movers in season is worse than a few days off the Buy Box.

Margin discipline

Some Buy Box wins aren't worth winning. The model had to know when to step away.

PRINCIPLE · *The Buy Box is won by the right price at the right time — not the lowest price all the time.*

Three signals fed one decision: re-price, hold, or step away.

I built a dynamic pricing and inventory gating model that combined three inputs and produced a per-SKU action recommendation each cycle.

Price elasticity

WHAT WE DID

- Estimated per-SKU sensitivity from historical price/sales data
- Distinguished elastic SKUs (price moves volume) from inelastic
- Bounded recommended price moves to defensible ranges

ROLE: *Tells you how much room the model has to move price without sacrificing volume.*

Competitor price monitoring

WHAT WE DID

- Continuous scrape of competitor offers per SKU
- Tracked Buy Box winner's price + offer attributes
- Detected when competitors raised, dropped, or stocked out

ROLE: *The market signal — there's no Buy Box without knowing what other sellers are doing.*

Safety-stock rules

WHAT WE DID

- Inventory thresholds per SKU based on lead time and demand
- Above-threshold: full price flexibility for Buy Box wins
- Below threshold: restricted discounting — protect supply

ROLE: *Prevents winning the Buy Box on a fast-mover the day before stocking out.*

From competitor scrape to repriced offer — every cycle.

Monitor
Competitor prices + Buy Box winners

Score
Elasticity + inventory state

Decide
Reprice / hold / step away

Update
Push new prices to Amazon

Measure
Buy Box occupancy + margin tracked

Margin floors as guardrails
Pricing engine never recommended below pre-set per-SKU margin floors. The model could lose a Buy Box rather than break a margin commitment.

Continuous re-evaluation
Decisions refreshed on cadence — the moment a competitor moved or an inventory threshold tripped, the model reconsidered.

DECISION PRINCIPLE · *Reprice, hold, or step away — three choices per SKU per cycle, each with explicit logic behind it.*

Buy Box occupancy up — and margin held.

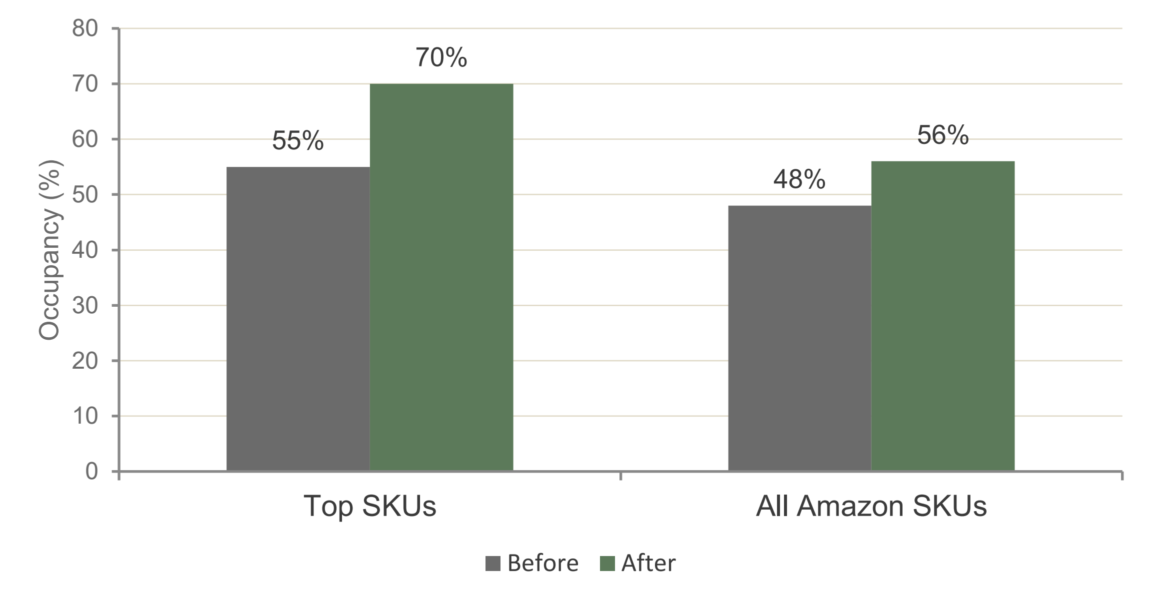
HOW WE MEASURED

Tracked Buy Box occupancy share per SKU before and after the engine went live, alongside realized GMV and contribution margin to confirm the wins didn't come from giving up margin.

WHAT MOVED

- Buy Box occupancy on top-selling SKUs
- GMV across the Amazon channel
- Contribution margin (the guardrail)
- Out-of-stock incidents during peak periods

BUY BOX OCCUPANCY — TOP-SELLING SKUs



Illustrative — directional values shown for confidentiality.

PROJECT 02 IMPACT

+15% Buy Box occupancy on top-selling SKUs — boosting GMV with margin protected by inventory & floor rules.

Two levers, one growth engine.

These weren't isolated wins — segmentation worked the demand side, Buy Box worked the supply side, and together they covered the full commercial loop.

DEMAND-SIDE · SEGMENTATION

Identified who to talk to

Customers grouped by behavior — premium, deal-driven, lapsing, one-time.

Defined what to offer

Tailored promotions, product mixes, and email cadences per segment.

Lifted DTC conversion + AOV

Same traffic, more revenue per visitor — more value extracted from existing demand.

SUPPLY-SIDE · BUY BOX

Captured marketplace position

Won the Amazon default "Add to Cart" position more often — directly drove GMV.

Protected margin & inventory

Floors and gating prevented winning Buy Box at a loss or stocking out of fast-movers.

Compounded with segmentation

Higher Buy Box occupancy meant the customer base segmented in P01 grew larger and faster.

The scoreboard.

CONVERSION RATE LIFT (DTC)

+15%

from segmentation-driven targeting

AVERAGE ORDER VALUE

+5–7%

tailored offers & cross-sell per segment

BUY BOX OCCUPANCY (TOP SKUs)

+15%

from dynamic pricing + inventory gating

OF SKUs MANAGED

Thousands

across Amazon, Walmart, and DTC channels

QUALITATIVE WINS

Marketing campaigns shifted from broad to targeted · Pricing decisions moved from manual repricing to systematic cycles · Margin held while occupancy and conversion both grew.

What worked, what I'd extend next.

WHAT WORKED

Picking k for usability, not silhouette

A statistically optimal 12-cluster solution that Marketing couldn't action would have been worthless. Fewer, clearer segments won.

Margin floors as Buy Box guardrails

Refusing to chase Buy Box wins below margin floors meant the gains were real, not borrowed from the P&L.

Owning data through deployment

Both projects shipped because I owned the path from raw transactions through to Marketing campaigns and pricing API calls.

WHAT I'D EXTEND NEXT

Move from RFM to behavioral embeddings

Hand-crafted RFM features beat nothing, but learned customer embeddings from sequence models would capture richer patterns at scale.

Causal uplift on segmented offers

Segmentation tells you who's likely to buy. Uplift modeling tells you who's persuadable — a sharper signal for offer targeting.

Reinforcement learning on pricing

Rules + elasticity + safety stock works. RL with margin and Buy Box reward signals could push further in markets where the rules space gets too complex.

Two analytics workstreams, one commercial story.

01

Build for the action, not the artifact.

k-means is interesting; tailored offer playbooks are revenue. The methodology earns its keep when it changes what someone does on Monday morning.

02

Pair demand-side and supply-side levers.

Segmentation alone wouldn't have moved Amazon GMV. Pricing alone wouldn't have lifted DTC AOV. Together they covered the full commercial loop.

03

Margin discipline isn't a constraint — it's the win condition.

Both projects could have produced bigger headline numbers by giving up margin. They didn't, and that's why the gains stuck.

Thank you · Questions?